PRESENTED BY
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GOLD AS AN INVESTMENT: HOW DOES IT BEHAVE IN DIFFERENT ECONOMIC CONDITIONS?

DATA ANALYSIS OF GOLD, INFLATION, GDP, AND OTHER METALS

Motivation

Why study gold?

Gold is widely viewed as a **safe-haven asset**. During periods of economic uncertainty such as high inflation and recessions, many investors turn to gold to protect their wealth.

However, many common beliefs about gold are based on **intuition or financial media narratives**, rather than clear data analysis.

By analyzing historical data on inflation, GDP growth, and other metal prices, this study provides evidence-based insights to help consumers **make more informed investment decisions.**

Who Is Our Audience?

This analysis is designed for non-expert investors

1/ individual investors deciding whether to include gold in their portfolio

2/ households concerned about inflation eroding purchasing power

3/ people seeking stable assets during economic uncertainty

These groups often hear that gold is a "safe investment," but may not know when or why it actually performs well.

Research Questions

Does gold perform better during **high inflation**?

Does gold perform better during **low GDP growth**?

How does gold compare with **silver** and **copper**?

Data Used In This Analysis

Macroeconomic Data

GDP Growth
Inflation Rate

Metal Price Data

Gold Price
Silver Price
Copper Price

Time Range

1960 – 2025

Sources

API (macroeconomic data):
https://www.alphavantage.co/documentation/
Datafile (metal price)
https://www.macrotrends.net/1333/historical-gold-prices-100-year-chart

Data Preparation

Data Quality Check

Missing values
Duplicated values
Oddities
Data types

Merge

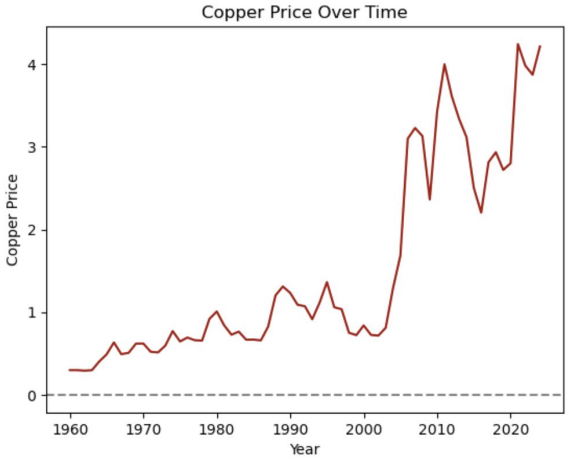
Match time frequencies
Merge all the datasets by year

Feature Engineering

gdp_growth	annual GDP change (%)
inflation_change	annual change of inflation (%)
gold_change_pct	annual gold return (%)
silver_change_pct	annual silver return (%)
copper_change_pct	annual copper return (%)
gold_inflation_gap	annual gold return - inflation
gold_gdp_gap	annual gold return - gdp growth

EXPLORATORY DATA ANALYSIS

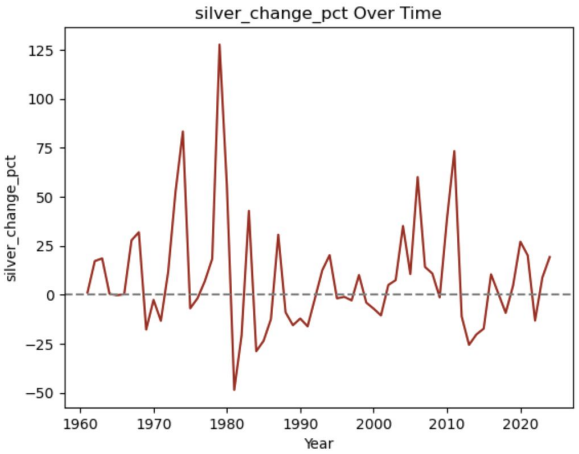
EDA: Initial Patterns in the Data

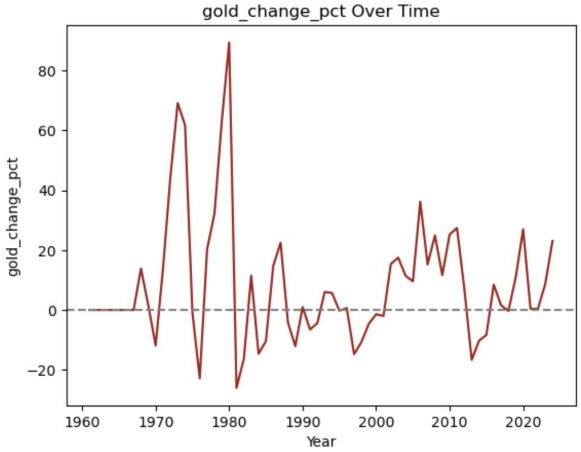


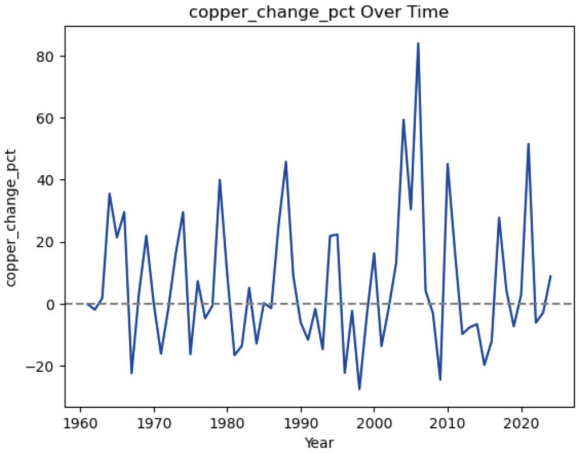
What the charts show

These charts plot the **historical trends** of gold prices, silver prices, copper prices over time.
Metal prices generally **trend upward**, reflecting increases in commodity prices and global demand.

EDA: Initial Patterns in the Data





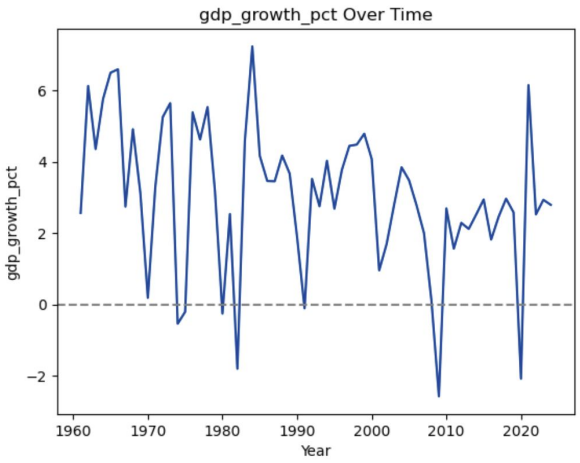


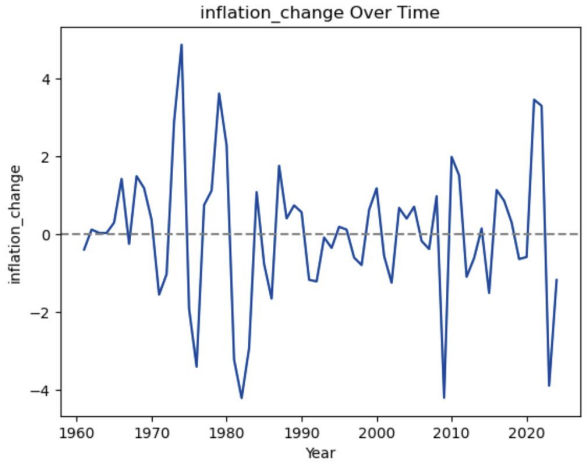
What the charts show

These charts plot how silver's, gold's, and copper's **year-to-year price changes** fluctuate across time.

Gold and silver often move together, suggesting that both metals may be influenced by similar financial market forces, while copper appears to follow a different pattern.

EDA: Initial Patterns in the Data

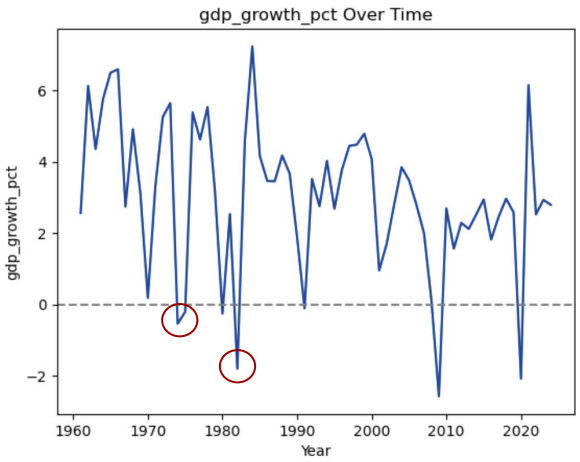


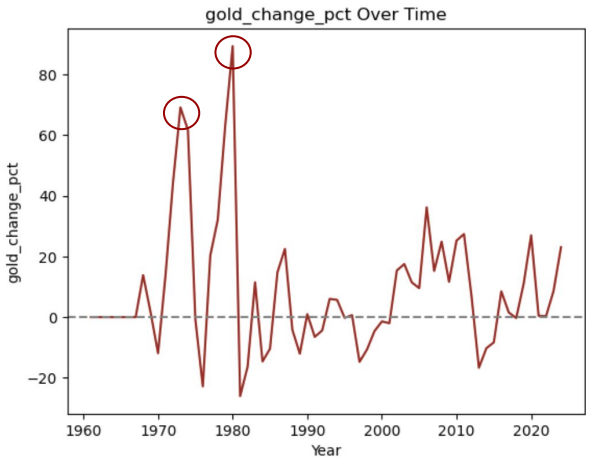


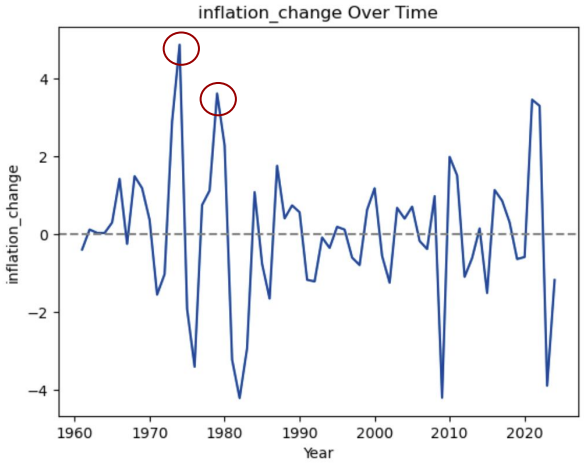
What the charts show

These charts plot the **year-to-year GDP and inflation changes** over time.
Periods of high inflation and economic slowdowns are clearly visible in the macroeconomic data.

EDA: Initial Patterns in the Data







What the charts show

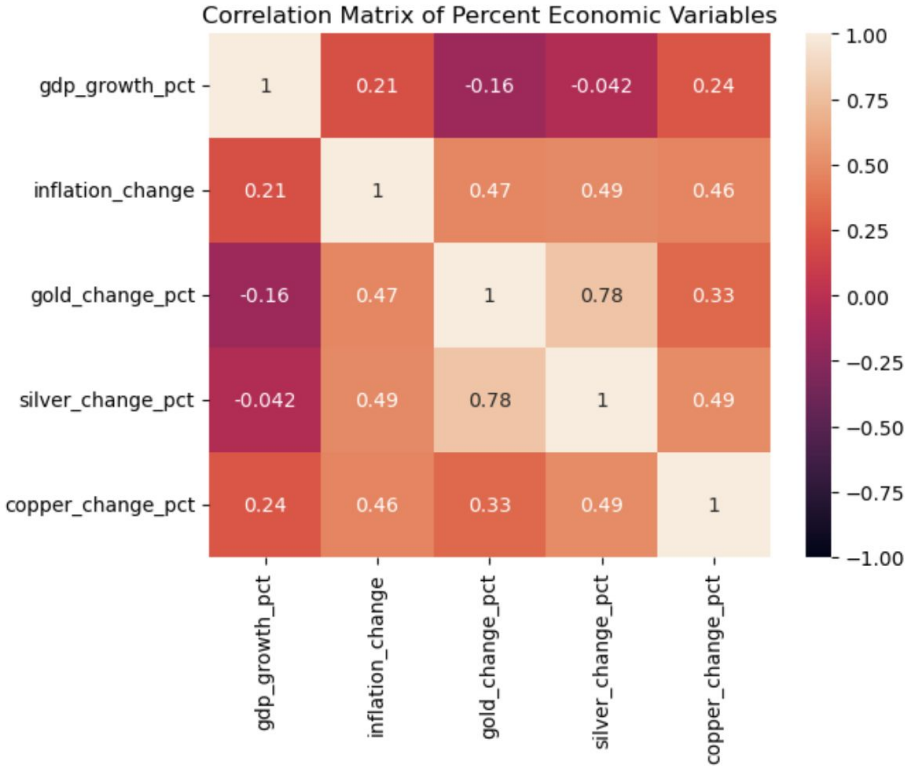
When combining the GDP, inflation fluctuations with year-to-year gold price change chart, we can find that **Gold price spikes appear during periods of economic instability** (high inflations/low gdp growth years).

What the heatmap shows

The heatmap also indirectly demonstrate the pattern that **gold and silver** tend to **move more closely together** than gold and copper. Gold prices may be influenced by **macroeconomic conditions** such as inflation and economic growth.

Next Step

However, visual inspection alone cannot determine whether these relationships are systematic or just coincidental.
To investigate this further, the next sections analyze the data more formally using SQL queries, subgroup comparisons, and regression models.



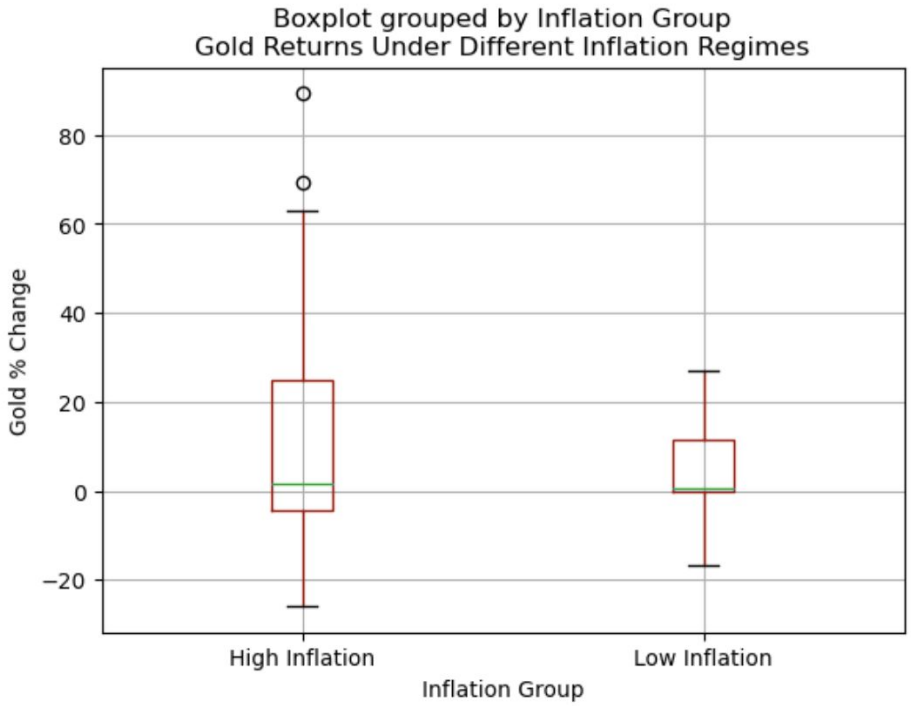
GOLD & INFLATION

Gold return in high- and low-inflation years

One of the main reasons non-expert consumers consider gold is the belief that it protects purchasing power when inflation rises.
To test this idea, we **compare gold returns** in high- and low-inflation years (subgroup analysis).

Boxplot of gold returns by GDP group

The overall distribution of gold returns is **higher** in high-inflation years.
Several of gold's **strongest positive-return years** occur in high-inflation periods.



Top gold-return years

Among the **top 10 gold return** years, **8/10** occurred during high inflation periods, suggesting that gold tends to perform especially well when inflation is high.

Purchasing power interpretation

To take a step further, we use **gold_inflaiton_gap** that measures whether gold merely rose in price or actually **outpaced inflation** itself.
Results show that during high-inflation years, gold **increased more than inflation**, meaning it preserved or increased purchasing power.

	year	inflation_group	inflation_rate	gdp_growth_group	gdp_growth_pct	gold_change_pct
0	1980	High Inflation	13.549202	Low GDP Growth	-0.256776	89.348859
1	1973	High Inflation	6.177760	High GDP Growth	5.645680	69.136725
2	1979	High Inflation	11.254471	High GDP Growth	3.165988	63.100506
3	1974	High Inflation	11.054805	Low GDP Growth	-0.540550	61.753304
4	1972	High Inflation	3.272278	High GDP Growth	5.255502	44.255492
5	2006	High Inflation	3.225944	Low GDP Growth	2.784540	36.206209
6	1978	High Inflation	7.630964	High GDP Growth	5.535206	32.122077
7	2011	High Inflation	3.156842	Low GDP Growth	1.564407	27.398605
8	2020	Low Inflation	1.233584	Low GDP Growth	-2.081376	26.992511
9	2010	Low Inflation	1.640043	Low GDP Growth	2.695193	25.248585

	year	inflation_group	inflation_rate	gold_price	gold_change_pct	gold_inflation_gap
0	1980	High Inflation	13.549202	607.762500	89.348859	75.799657
1	1979	High Inflation	11.254471	320.975000	63.100506	51.846035
2	1974	High Inflation	11.054805	161.679167	61.753304	50.698499
3	1978	High Inflation	7.630964	196.795833	32.122077	24.491113
4	1977	High Inflation	6.501684	148.950000	20.323124	13.821440
5	1973	High Inflation	6.177760	99.954167	69.136725	62.958965

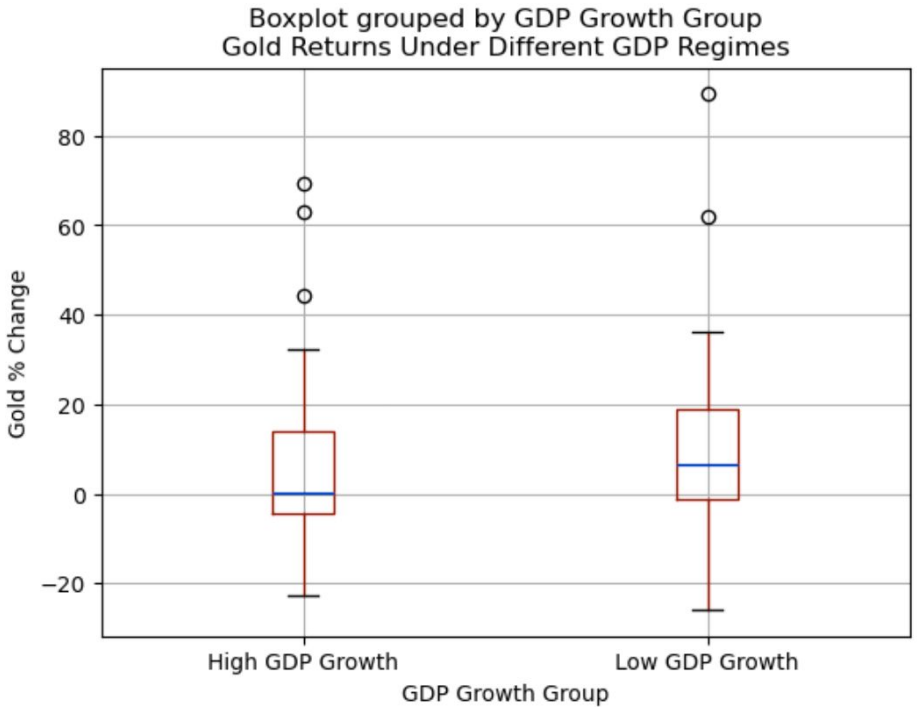
GOLD & GDP

Gold return in high- and low-GDP-growth years

Another main reasons non-expert consumers consider gold is the belief that it is a **defensive asset** when economic growth weakens.
To test this idea, we **compare gold returns** in high- and low-GDP years (subgroup analysis).

Boxplot of gold returns by GDP growth group

The median gold return appears **higher** during low GDP growth periods.
Several of gold's **strongest positive-return years** occur in low-GDP periods.
Gold does **not always** perform well during weak economic growth. Some negative returns appear in both groups, suggesting that other factors may also affect.



Top gold-return years

Among the **top 10 gold return** years, **6/10** occurred during low GDP periods, suggesting that gold is more likely to perform especially well when GDP is low.

	year	inflation_group	inflation_rate	gdp_growth_group	gdp_growth_pct	gold_change_pct
0	1980	High Inflation	13.549202	Low GDP Growth	-0.256776	89.348859
1	1973	High Inflation	6.177760	High GDP Growth	5.645680	69.136725
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8	2020	Low Inflation	1.233584	Low GDP Growth	-2.081376	26.992511
9	2010	Low Inflation	1.640043	Low GDP Growth	2.695193	25.248585

Purchasing power interpretation

To take a step further, we use **gold_gdp_gap** that measures whether gold can **outpace GDP growth** itself. Results show that during low-GDP growth years, gold **increased more than GDP growth**, suggesting it may serve as a defensive asset during weaker economic conditions.

	year	gdp_growth_group	gdp_growth_pct	gold_change_pct	gold_gdp_gap
0	1980	Low GDP Growth	-0.256776	89.348859	89.605635
1	1974	Low GDP Growth	-0.540550	61.753304	62.293854
2	2006	Low GDP Growth	2.784540	36.206209	33.421669
3	2011	Low GDP Growth	1.564407	27.398605	25.834198
4	2020	Low GDP Growth	-2.081376	26.992511	29.073887
5	2010	Low GDP Growth	2.695193	25.248585	22.553392

GOLD & OTHER METALS

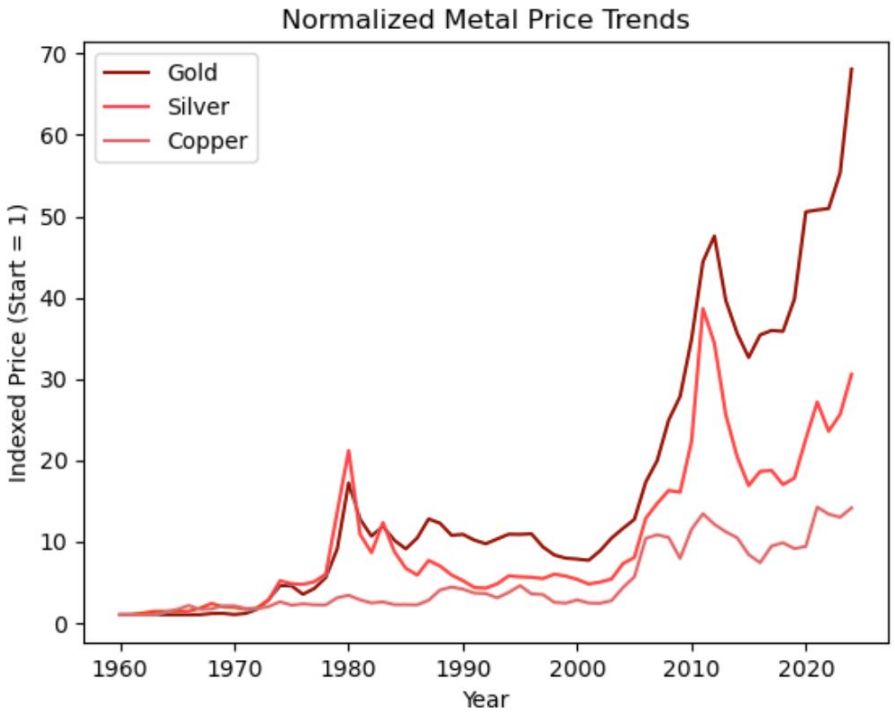
Gold, silver, & copper annual returns

Among the 65 years in our data, there are **22** years when gold's annual return exceeded **both** silver and copper.

Normalized Metal Price Chart

The normalization **removes the scale differences (price)** between metals, allowing us to compare their long-term growth trajectories.

From the chart:
Gold and silver often **move together.**
Copper **behaves differently** possibly due to industrial demand.

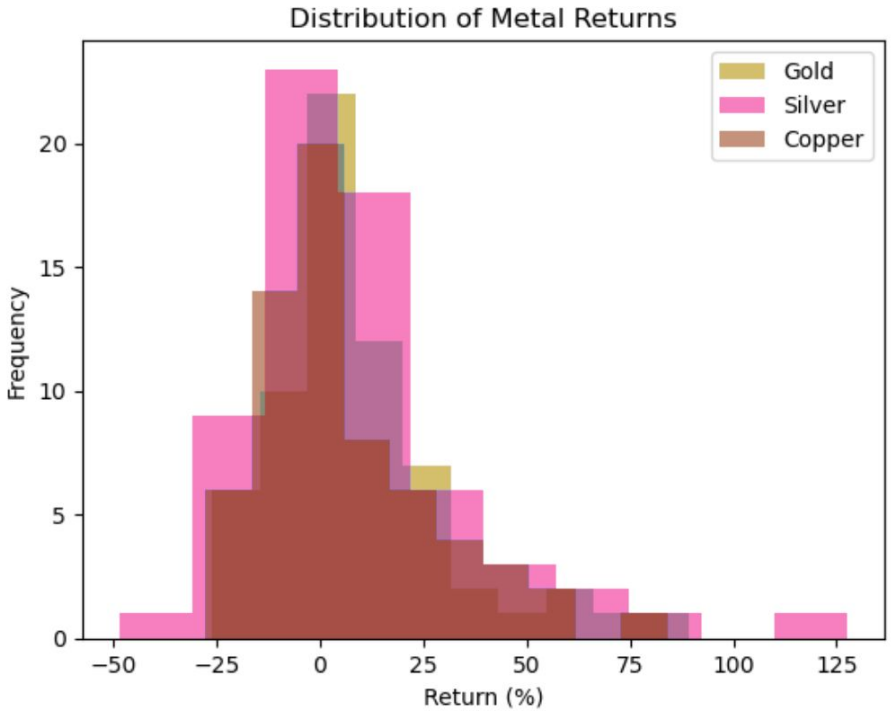


Metal stability (volatility)

Volatility refers to the **degree of variation** in an asset's returns over time, which investors often consider as **risk**. The volatility analysis reveals that **silver** has the **highest volatility** (28.55), and gold (21.46) is relatively stable compared to copper (21.67).

The histogram of distribution also shows:
Silver's distribution is the widest, while gold's distribution is more concentrated.

For investors, gold's relatively lower volatility and its tendency to perform well during macroeconomic stress suggest that gold may serve as a defensive asset.



REGRESSION ANALYSIS

Model 1

Gold Returns & Inflation, GDP Growth

Explanation

Model 1 examines whether **macroeconomic conditions** alone help explain changes in gold returns.

This model tests the hypothesis that gold behaves as a **macro hedge asset** that responds to changes in inflation or economic growth.

Results & Implications

coef(GDP growth) = -2.7873, p(GDP growth) = .018
coef(inflation change) = 6.4684, p(inflation) < .001
R^2 = 0.29

Model 1 supports the earlier descriptive evidence that gold tends to perform **better when inflation becomes more intense** and tends to perform **less strongly when the economy grows faster.**

Model 2

Gold Returns & Silver, Copper

Explanation

Model 2 examines whether gold returns are more closely related to the **returns of other metals**.

This model helps answer whether gold **behaves more like**:

- silver, another precious metal
- copper, an industrial metal

Results & Implications

coef(silver return) = .6148, p(silver return) < .001
coef(copper return) = -.0698, p(inflation) = .585
R^2 = 0.619

Model 2 supports the earlier evidence that gold **moves much more closely with silver** than with copper, suggesting gold behaves more like a precious metal influenced by investor demands.

Model 3

Gold Returns & Inflation, GDP Growth, Silver, Copper

Explanation

Model 3 examines when macroeconomic conditions and metal-market variables are **all considered together**, which ones still matter the most for gold returns.

This model helps determine whether the earlier results remain meaningful **when all major variables are included at the same time**.

Results & Implications

coef(GDP growth) = -.9504, p(GDP growth) = .256
coef(inflation change) = 1.4600, p(inflation) < .024
coef(silver return) = .5536, p(silver return) < .001
coef(copper return) = -.0198, p(inflation) = .789
R^2 = 0.662

Model 3 again confirms that **inflation** and **silver** returns remain the most important factors when everything is included together. Copper remains less relevant.

CONCLUSION

Key Findings

1. **Inflation Matters**
   Both the descriptive analysis and regression results show that inflation has a positive relationship with gold returns.

2. **Gold often performs well when growth slows**
   Both the descriptive analysis and regression results suggest that stronger economic growth is associated with weaker gold performance.

3. **Gold moves closely with silver**
   The metal comparison analysis and regression results both show a strong relationship between gold and silver.

4. **Copper has little influence on gold**
   Copper returns showed a very weak and statistically insignificant relationship with gold returns in the regression models.

Implications

For **non-expert consumers and investors**

Gold may be mostly a useful **safe asset** during periods of rising inflation and periods of economic uncertainty or slow growth.

Gold may also be a **diverse asset** influenced by investor demands that behaves differently from industrial commodities.

Limitations

1/ The dataset only includes **a limited set of variables**. Gold prices may also be influenced by other factors
2/ The analysis uses **annual data**, which may hide **short-term** fluctuations in gold prices.
3/ Regression models identify statistical associations, but they do not necessarily prove **causal relationships**.

Future Research

1/ incorporate additional macroeconomic variables
2/ examine **real interest rates**, which are often linked to gold prices using **higher-frequency data**
3/ apply **time-series models or machine learning methods**.

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THANK YOU